TN Vaazhga — Complete Launch & Build Plan

A step-by-step guide to build, launch, operate, secure, and monetize a Tamil-first AI citizen platform for Tamil Nadu.

Prepared for: Digital marketing agency owner / developer, Tamil Nadu

Product type: Mobile-first AI web platform + future WhatsApp assistant

Core promise: Not just answers — a Tamil Nadu action plan for schemes, documents, jobs, education, agriculture, and local services.

1. Product Vision

Product name: TN Vaazhga (working name)

Tagline: Not just answers. Your Tamil Nadu action plan.

TN Vaazhga helps citizens discover eligible government schemes, understand required documents, find nearby help centres, track deadlines, and receive simple Tamil guidance. It is a guidance layer above official government portals—not a replacement for them.

Why it is different from ChatGPT/Gemini:

- Curated Tamil Nadu scheme database with official source and last-verified date.
- Rules-based eligibility matching, not only free-text AI answers.
- Document Doctor: missing-document checklist and renewal guidance.
- Deadline Radar: application, renewal, exam, and scholarship reminders.
- District-aware help-centre locator.
- Tamil-first interface with voice input and optional voice replies.
- Database-first + cache-first architecture to keep AI cost low.

2. Target Users

User group	Main need	Priority features
Students	Scholarships, courses, exams, admission	Scholarship matcher, exam alerts, document checklist
Job seekers	Government exams and job eligibility	TNPSC/TNUSRB/TET eligibility, notifications, resume help
Women & families	Welfare schemes, maternity, safety	Scheme matcher, documents, helplines
Farmers	Subsidies, crop guidance, insurance	Crop schemes, weather, expert-verified guidance
Senior citizens	Pensions and renewals	Voice-first UI, large text, pension checker
Small business owners	Licenses, GST, MSME, loans	Business checklist, scheme guidance
General citizens	Certificates, bills, land records	Document Doctor, local services, official links

3. Core Product Loop

1. User creates a 30-second profile.
2. System matches relevant schemes using rules.
3. User receives schemes, missing documents, deadlines, and next steps.
4. User saves a Urimai Card, downloads a checklist, or shares via WhatsApp.
5. System sends reminders and improves recommendations over time.

4. Website Pages & Features

#	Page	Main features
1	Home	AI ask box, category cards, trending schemes, voice CTA
2	Government Schemes	Eligibility matcher, search, filters, scheme cards, Urimai Card
3	Certificates & Documents	Document Doctor, checklists, apply guidance, renewal reminders
4	Jobs & Exams	Exam matcher, notifications, syllabus summary, resume builder
5	Education & Scholarships	Course guidance, scholarship matcher, admission checklist
6	Agriculture	Crop schemes, subsidies, insurance, weather, pest guidance
7	Health	Insurance guidance, hospital locator, maternity/disability guidance
8	Women & Family Welfare	Women schemes, maternity benefits, helplines, SHG guidance
9	Senior Citizens & Pensions	Pension eligibility, renewals, voice-first interface
10	Business & Self-Employment	Shop/FSSAI/GST/MSME guidance, business checklists
11	Bills & Local Services	Property/water/electricity guidance, official payment links
12	Emergency & Help Centres	Nearest CSC/e-Sevai, hospitals, police, emergency numbers

5. UI/UX Design System

Design principle: One primary task per screen, large buttons, Tamil-first content, progressive disclosure, and clear official-source trust signals.

- Mobile-first layout; desktop is a wider version of the mobile flow.
- Default language: Tamil; English toggle in header.
- Font: Noto Sans Tamil or Mukta Malar; body size 17–19px on mobile.
- Primary color: deep blue; accent: saffron/yellow; status colors: green, yellow, red.
- Buttons minimum 48px height; sufficient spacing for touch targets.
- Every scheme answer shows official link, department, and last-verified date.
- No login required for basic questions; login only for saving profile/checklists/reminders.

Home screen wireframe:

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Welcome + Ask in Tamil + Category cards + Trending schemes + Bottom navigation
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Scheme page wireframe:

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Scheme name + eligibility + documents + apply method + official link + Save/Share/Check eligibility
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Eligibility wizard: Ask one question at a time: age, district, gender, occupation, income, need. Show match percentage and next action.

6. Technical Architecture

Use a database-first and cache-first architecture. Gemini is used only for language understanding, Tamil formatting, and complex queries—not for every button click.

Layer	Recommended technology	Purpose
Frontend	Next.js + Tailwind CSS	Fast, SEO-friendly, mobile-first UI
Backend	Next.js API routes or Node.js	Business logic, secure API layer
Database	Supabase PostgreSQL	Users, schemes, saved items, reminders
Authentication	Supabase Auth	Email/OTP/Google login
AI	Gemini 2.5 Flash	Tamil understanding, replies, summaries
Cache	Upstash Redis	FAQ cache, rate limits, session data
Search	Postgres full-text search	Scheme/document/exam search
Maps	Leaflet + OpenStreetMap	Nearest centres, low-cost maps
File storage	Supabase Storage	Public PDFs, images, admin assets
Email	Resend / SMTP	OTP, alerts, admin notifications
Analytics	PostHog / GA4	User behaviour and funnel tracking
Error tracking	Sentry free tier	Bug monitoring

7. Database Schema — MVP

Table	Important fields
users	id, email, phone, name, district, age, gender, occupation, income_range, language, created_at
schemes	id, slug, title_ta, title_en, department, category, eligibility_json, documents_json, apply_method, official_url, distric
documents	id, name_ta, name_en, issuing_authority, process, cost, duration, official_url

Table	Important fields
user_documents	id, user_id, document_id, status, expiry_date, notes
saved_schemes	id, user_id, scheme_id, match_score, status, created_at
reminders	id, user_id, title, due_date, channel, status
chat_messages	id, user_id, role, content, intent, created_at
faq_cache	id, question_hash, normalized_question, answer, source_url, hit_count, created_at
scheme_updates	id, scheme_id, change_type, old_value, new_value, verified_by, verified_at
help_centres	id, name, type, district, address, lat, lng, phone, hours, services

8. AI Cost-Control Workflow

Every request follows this sequence:

- Clean and normalize the Tamil input.
- Check Redis semantic/FAQ cache.
- If cached, return instantly at zero Gemini cost.
- If not cached, search the scheme database using filters/keywords.
- If a confident match exists, use rules engine and ask Gemini only to format the Tamil reply.
- If ambiguous, send limited context to Gemini Flash.
- Store the response in cache with a TTL.
- Log token usage, latency, cache hit, and model version.

Cost rules: Keep prompts under 3,000–5,000 tokens; cap normal answers at 250–400 output tokens; avoid sending full database context; use streaming; cache aggressively; reserve large-model/tool usage for unclear queries.

9. Fast Voice Experience

Target: First Tamil voice response within 1–2 seconds after the user stops speaking.

Stage	Target
Speech-end detection	200–400 ms
Streaming STT first result	300–800 ms
Gemini first token	500–1,200 ms
First TTS audio	1–2 seconds
Full short answer	3–6 seconds

V1 voice approach: Browser speech input + on-demand Tamil TTS button. This is cheaper and simpler.
V2 voice approach: Streaming STT → Gemini streaming → streaming TTS over WebSocket, so the first sentence plays while the rest is generated.

- Use streaming STT, streaming LLM output, and streaming TTS.
- Keep prompts short and send only top matching scheme context.
- Limit normal output to 250–400 tokens.
- Cache common questions and answers.
- Use Opus/low-latency streaming audio where supported.
- Show live transcript and “listening/checking/speaking” states immediately.

10. Website Performance Checklist

Area	Action	Target
Core Web Vitals	Optimize LCP, INP, CLS	LCP < 2.5s, INP < 200ms, CLS < 0.1
Fonts	One Tamil font, preload, font-display swap	No layout shift
Images	WebP/AVIF, responsive sizes, lazy loading	Small page weight
JavaScript	Code splitting, avoid heavy libraries	Fast interaction
API	Pagination, selected fields, compression	Fast responses
Database	Indexes on district/category/scheme/title	Fast search
Cache	Redis for FAQ, scheme pages, location results	Instant repeat answers
Rendering	SSR/ISR for public scheme pages	SEO + fast first paint

11. Login & Security Plan

Login strategy: Basic scheme search and AI questions should work without login. Ask login only when the user saves schemes, creates a document wallet, sets reminders, or needs personalized history.

- V1 login: Google login + email OTP. Avoid complex password flows.
- Store minimal personal data: name, district, age band, occupation, language.
- Never store Aadhaar, PAN, bank credentials, passwords, or sensitive document images in V1.
- Use Supabase Row Level Security so users can access only their own records.
- Keep Gemini API key only in server environment variables; never expose it in frontend code.
- Rate-limit AI endpoints per IP/user; add CAPTCHA or proof-of-work only if abuse appears.
- Use HTTPS everywhere, secure cookies, CSRF protection, input validation, and audit logs.
- Display clear disclaimer: “This is not an official Tamil Nadu Government website.”

12. Content & Data Operations

- Start with 50–100 high-demand schemes, not every scheme.
- For each scheme, collect: official URL, department, eligibility, documents, apply method, fee, timeline, districts, last-verified date.
- Create Tamil and English versions for every scheme.
- Run weekly checks on high-traffic schemes and monthly checks on others.
- Allow users to report outdated information; admin verifies before publishing.
- Maintain a change log for every scheme update.
- Never present AI output as an official government decision.

13. Development Roadmap

Phase	Duration	Deliverables
Phase 0 — Planning	Week 1	Final scope, brand, domain setup, 50 priority schemes, wireframes
Phase 1 — MVP build	Weeks 2–5	Home, scheme search, eligibility matcher, document checklist, AI chat, Supabase auth
Phase 2 — Trust & growth	Weeks 6–7	Urimai Card, WhatsApp share, deadline reminders, help-centre locator, analytics
Phase 3 — Voice & scale	Weeks 8–10	Tamil voice input, TTS, performance optimization, admin panel
Phase 4 — Monetization	Weeks 11–12	Pro plan, B2B pilot, sponsored listings, WhatsApp automation

14. Launch Checklist

- Domain SSL, www/non-www redirect, favicon, OG image, meta tags.
- Tamil and English sitemap, robots.txt, Google Search Console, Bing Webmaster.
- Google Analytics 4 or PostHog, error tracking, uptime monitoring.
- Privacy policy, terms, AI disclaimer, contact/support page.
- Mobile testing on low-end Android and slow 3G/4G.
- Accessibility check: contrast, alt text, keyboard navigation, screen-reader labels.
- Load test AI endpoint and set spending/rate limits.
- Seed content: 50 schemes, 100 FAQs, 20 document guides, emergency contacts.
- Beta test with 20–50 real users; fix top 10 confusions.
- Launch campaign: Instagram reels, WhatsApp groups, local college/student communities, LinkedIn build-in-public posts.

15. Pricing & Monetization

Keep the core scheme guidance free to build trust and traffic. Monetize convenience, B2B tools, and business services—not basic citizen rights information.

Plan	Audience	Monthly price	Includes
Free	All citizens	₹0	Basic Q&A, 3 scheme checks/day, public checklists, official links
Plus	Regular users	₹49/month or ₹499/year	Unlimited scheme checks, saved profile, document wallet, reminders
Pro	Power users/students	₹99/month or ₹999/year	Priority AI, exam/job alerts, resume tools, advanced planner
CSC / Agent	e-Sevai agents, CSC centres	₹999/month	Multi-user assistance, document templates, lead inbox, branding
Business	Local shops, clinics, colleges	₹1,999–₹4,999/month	AI assistant, WhatsApp automation, lead capture, sponsored visibility

16. Revenue Projection Example

Scenario	Free users	Paid users	CSC/B2B	Estimated monthly revenue
Launch month	500	10	0	₹990
Month 3	3,000	75	5	₹12,430
Month 6	10,000	250	20	₹44,980
Month 12	30,000	750	50	₹1,27,475

Assumption: average paid revenue ₹99–₹199/user and CSC/B2B average ₹1,999/month. These are illustrative targets, not guarantees.

17. Marketing Launch Plan

- Build in public on LinkedIn: daily development, UI, and user-feedback updates.
- Instagram reels in Tamil: “Government scheme details in 30 seconds”, “Documents you need”, “Free AI helper for Tamil Nadu”.
- WhatsApp shareable Urimai Cards with branded footer link.
- College partnerships for scholarship/job guidance.
- CSC agent partnership program for local distribution.
- Local SEO pages: “Tamil Nadu government schemes in Tamil”, “income certificate documents Tamil”, “TNPSC eligibility Tamil”.
- Press/creator outreach after first 1,000 users.

18. First 30 Days Action Plan

Week	Focus	Output
Week 1	Finalize scope, database schema, 50 schemes, wireframes	Product spec + Figma-style wireframes
Week 2	Next.js setup, Supabase schema, auth, basic UI	Working skeleton
Week 3	Scheme search, eligibility rules, AI chat endpoint	Core feature working
Week 4	Document Doctor, saved schemes, Urimai Card, WhatsApp share	Beta-ready product
Week 5	Performance, security, analytics, content seeding	Public beta launch

19. Risks & Mitigation

Risk	Mitigation
Outdated or wrong scheme information	Official source link, last-verified date, user correction reports, admin review
High Gemini cost	Cache-first design, rules engine, short prompts, output limits, usage dashboard
User trust issues	Clear non-official disclaimer, no sensitive data storage, transparent sources
Slow voice replies	Streaming pipeline, short responses, cache, on-demand TTS in V1
Low retention	Deadline reminders, document wallet, personalized scheme matching
Content workload	Start with 50 schemes; use templates and weekly update workflow

20. Final Recommendation

Launch a focused MVP with five sections: Government Schemes, Certificates & Documents, Education & Scholarships, Jobs & Exams, and Emergency Help. Use Supabase + Next.js + Gemini Flash + Upstash Redis. Keep basic guidance free, charge only for convenience and B2B features, and grow content gradually after real user feedback.

Next immediate action: Create the Supabase schema, collect the first 50 schemes, and build the homepage + scheme detail page + eligibility wizard.